

ET AI HACKATHON 2026 · PROBLEM STATEMENT 4

PRAMAAN

EPC Deviation Intelligence for Hyperscale Data Centres

Spec-to-Site Deviation Sentinel + Commissioning Risk Twin

*Pramaan (प्रमाण) — Sanskrit for **proof**. Named on purpose: every number here is reproducible.*

50

Deviations Caught

1,024

Weeks Saved

1.000

Real-doc Recall

12

Projects

THE PROBLEM

Deviations Hide in Thousands of Pages

In a **40 MW Tier IV** data centre build, the design basis, vendor submittals, and governing standards live in different systems, reviewed by different people.

Today

Week 16–44

Deviations surface during commissioning — rework, schedule delays, cost overruns

With Pramaan

Week 11

All 14 caught the day the submittal was uploaded — under 5 minutes

WHAT GOES WRONG

14 Critical Deviations in One Project

DEVIATION	SPEC SAYS	VENDOR SUBMITTED	IMPACT	LEAD
UPS battery runtime	10 min	7 min	Tier IV fault tolerance broken	27w
Generator fuel autonomy	24 h	12 h	Design-duration outage failure	30w
Cooling redundancy	N+2	N+1	No concurrent maint. tolerance	28w
BMS monitoring	Dual	Single	Single point of failure	33w
Generator start time	10 s	15 s	Battery depletion risk	23w
BMS alarm coverage	Complete	Missing leak	IST-14 alarm test will fail	29w
Cable fire rating	CMP	CMR	NFPA 75 violation in plenum	11w

Showing 7 of 14 · Total lead time: **267 weeks** (5+ years)

THE HEADLINE

267 weeks

of total lead time saved across 14 deviations

*Praaman caught a critical BMS monitoring single-point-of-failure **33 weeks** before it would have failed the full-facility failover drill. That's the difference between a one-line email and a seven-figure schedule slip.*

THE SOLUTION

5 AI Agents, One Pipeline

LangGraph orchestrator cross-references every requirement against every submittal against every governing standard



LangGraph StateGraph · Conditional routing · TypedDict state · Validation gate

DETECTION ACCURACY

Honest Detection Numbers

The synthetic portfolio below scores 1.000 *by construction* — a plumbing / breadth check, not a flex.

1.000

Precision

1.000

Recall

1.000

F1 Score

0

False Positives

1.000

Cx Prediction

17

Cx Tests Mapped

160+

Automated Tests

The number that matters: 11 sourced real datasheet pairs — 27 deviations (recall 1.000), 0 false positives, and a self-scored ≈ 0.9 on one contested ASHRAE case (live-verified, gemini-2.5-flash). We report ≈ 0.9 , not a suspicious 1.000.

MULTI-PROJECT GENERALISATION

12 Projects · 11 Countries · 6 Tier Standards

Synthetic breadth test — deviations are seeded, so F1 is 1.000 *by construction*. The real-world proof is the sourced-datasheet pairs, not this grid.

Meghdoot Tier IV · Mumbai · 40MW 14 devs · 267w saved F1 = 1.000	Vajra Tier III · Pune · 20MW 4 devs · 95w saved F1 = 1.000	Nordic Edge EN 50600 · Oslo · 10MW 5 devs · 113w saved F1 = 1.000	Sahara Tier II · Dubai · 5MW 3 devs · 38w saved F1 = 1.000	Cascade Tier IV · Oregon · 30MW 4 devs · 80w saved F1 = 1.000	Yangtze GB 50174 · Shanghai · 50MW 3 devs · 98w saved F1 = 1.000
Athena EN 50600 · Frankfurt · 15MW 4 devs · 80w saved F1 = 1.000	Sakura JEITA · Tokyo · 25MW 3 devs · 58w saved F1 = 1.000	Outback Tier III · Sydney · 8MW 3 devs · 53w saved F1 = 1.000	Maple Tier III · Toronto · 12MW 3 devs · 58w saved F1 = 1.000	Pampas Tier II · São Paulo · 6MW 2 devs · 34w saved F1 = 1.000	Thames Tier IV · London · 35MW 2 devs · 50w saved F1 = 1.000

1,024

Total Weeks Saved

256

Total MW

25+

Standards Covered

LIVE DASHBOARD

22-Section Dashboard · 27 Components

AWARENESS

- ▶ Hero Intro & Problem
- ▶ Sentinel Alert
- ▶ 267-Week Counter
- ▶ Before / After Toggle

INTELLIGENCE

- ▶ System Health Grid
- ▶ Deviation Register
- ▶ Risk Matrix
- ▶ Document Diff Viewer
- ▶ Cx Risk Twin (Gantt)

ACTION

- ▶ Live PDF Analysis
- ▶ RFI Copilot (RAG)
- ▶ ROI Calculator
- ▶ Export Evidence Pack
- ▶ Multi-Project Eval

Next.js 15 · Dark theme · Scroll-reveal animations · Responsive · Streaming AI

EVAL HARNESS

Reproducibility checks + proof on real documents

Paths 1–2 are seeded-corpus integrity checks (1.000 *by construction*, not detection skill); Path 3 — real, unseen datasheets — is the credibility signal.

Path 1: Structured

Pre-extracted triples vs ground truth

F1 = 1.000

Data integrity proof

Path 2: Text-Based

Raw markdown → regex extraction

F1 = 1.000

Independent input path

Path 3: LLM Agent

Real, unseen datasheets → LLM reasoning → deviations

Recall 1.000

27/27 hard · 0 FP · ~0.9 precision (live-verified)

`make eval-all` — two reproducibility checks + the real-document proof

DON'T TRUST US — RUN IT

Every Number on These Slides Reproduces Offline

One command. No API key. A judge's laptop is enough.

```
git clone · make verify
```

- ▶ **448 tests** — API, agents, security, resilience, real pairs
- ▶ **3 eval harnesses** — every headline metric regenerated
- ▶ **Type-check + production build** — CI green, publicly visible
- ▶ **No-key real-datasheet harness** — `eval/real_pairs_offline.py`

We surveyed ~70 public repos across every track of this hackathon: **zero CI pipelines, zero eval harnesses, one minimal test suite.**
Claims are easy. **Pramaan means proof.**

SCALE STORY

Demo → Enterprise

Current Demo

- ▶ 12 projects, 11 countries
- ▶ 81 systems across portfolio
- ▶ 203 requirements tracked
- ▶ 50 deviations detected
- ▶ TF-IDF retriever
- ▶ Synchronous agents

At Enterprise Scale

- ▶ Hundreds of concurrent projects
- ▶ 500+ systems per project
- ▶ 14,000+ line items per project
- ▶ Continuous monitoring pipeline
- ▶ pgvector / Qdrant vector store
- ▶ LangGraph async + task queue

HACKATHON RUBRIC

How Pramaan Maps to Every Dimension

Innovation

Beyond AI submittal review: predicts which commissioning test each deviation fails and how many weeks early. Cross-references spec + submittal + standard with a citation chain. 5 agents, LangGraph orchestration.

Business Impact

1,024 weeks of early detection across 50 findings in 12 projects. Interactive ROI calculator shows ₹1,788 lakhs avoided on ₹800 Cr project.

Technical Excellence

Synthetic portfolio is 1,000 by construction (breadth check); the real signal is 15 sourced datasheet pairs — 27 deviations (recall 1,000), 0 false positives, self-scored ≈ 0.9 on one contested case. 435 tests. 28 endpoints. Independent text + real-LLM evaluation.

Scalability & UX

22-section dashboard with 60-second demo narrative. Multi-project eval. Streaming AI. PDF upload. Evidence export. Responsive design.

TECH STACK

Built With

LLM

Gemini 2.5 Flash

ORCHESTRATION

LangGraph

BACKEND

FastAPI + SSE

FRONTEND

Next.js 15 + React 19

PDF

pdfplumber + PyMuPDF

RETRIEVAL

TF-IDF → pgvector

EVAL

Triple Harness

SCRAPING

Firecrawl + Playwright

DEPLOY

Vercel + Render

BUSINESS IMPACT

The ROI That Sells Itself

₹800 Cr

Project Value

₹1,788L

Rework Avoided

< 30

Payback (Days)

Every deviation carries `lead_time_weeks`.
The metric that wins is **time** — not accuracy alone.

THANK YOU

PRAMAAN

50 deviations · 1,024 weeks saved · 12 projects · 11 countries

Real-doc: 0 false positives · self-scored ~0.9 · 448 tests

► Live Demo

API Docs

parth-tan.vercel.app · parth-3puc.onrender.com